25901180 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Core Course (CC) - MG4CCRBCA200 - ARTIFICIAL INTELLIGENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Identify the primary goal of a problem-solving agent for a problem scenario. [K] / [CO1]
2. What is an AND-OR graph? [K] / [CO2]
3. What is meant by optimality in artificial intelligence search techniques? [K] / [CO2]
4. Explain the procedure of forward chaining. [U] / [CO3]
5. What is meant by Traditional rule based expert system? [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Explain Wumpus World problem in detail. [U] / [CO1]
7. Describe the different nature of environments in Artificial Intelligence. [U] / [CO1]

8. Explain the concept of Rationality in AI. What makes an agent rational? [U] / [CO1]

9. Analyze the impact of heuristics and constraint propagation methods on the structure and complexity of CSP solving. [An] / [CO2]

10. Examine and interpret the roles of selection, crossover, and mutation in Genetic Algorithms in achieving evolutionary optimization. [U] / [CO2]

11. List the differences between forward chaining and backward chaining. [K] / [CO3]

12. Explain the concept of unsupervised learning. [U] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Describe MIN-MAX algorithm in detail with its principle, algorithm and methods to enhance performance along with an example. [A] / [CO2]

14. Explain the working of inference mechanisms in First Order Logic. [U] / [CO3]

15. What is meant by NLP? Explain the different core NLP Components? [U] / [CO4]

[15x2 = 30]